INCH-POUND
MIL-D-24693(SH)
23 November 1987

MILITARY SPECIFICATION

DEHYDRATOR, CONDENSER-FILTER LOW PRESSURE AIR, FOR SPECIAL APPLICATIONS

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements for an automatic condenser-filter type dehydrator for use in low pressure air systems. The condenser-filter dehydrator shall remove particulate matter, aerosols (see 6.5) and water vapor from influent low pressure air and furnish clean, dry air as specified herein for use in propulsion plant control systems, operating automatic combustion control and other special low pressure air applications. The condenser-filter dehydrator shall be designed for both continuous and intermittent operation. The word "dehydrator" will be used to denote the entire complex of equipment throughout this specification.

1.2 Classification.

1.2.1 Capacity. The nominal capacity (nominal effluent air flow rate) for the dehydrator shall be of the following classes as specified (see 6.2.1).

Class 1 - 100 standard cubic feet per minute (ft^3/min).
Class 2 - 200 ft^3/min.
Class 3 - Special, nominal capacity as specified.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
TT-E-490 - Enamel, Silicone Alkyd Copolymer, Semigloss (For Exterior and Interior Use).
TT-P-645 - Primer, Paint, Zinc Chromate, Alkyd Type.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-C-104 - Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915 - Cable and Cord, Electrical, for Shipboard Use, General Specification for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2212 - Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-S-8805 - Switches and Switch Assemblies, Sensitive and Push (Snap Action), General Specification for.
MIL-T-15005 - Tube, 70-30 and 90-10 Copper-Nickel Alloy, Condenser and Heat Exchanger.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-A-15303 - Audible Signals: Alarms, Bells, Buzzers, Horns, and Sirens, Electronic, Shipboard.
MIL-S-16032 - Switches and Detectors, Shipboard Alarm Systems.
MIL-I-18997 - Indicator, Pressure, Panel Mounted or Case Supported, General Specification.
MIL-I-18997/1 - Indicators, Pressure, Circular Dial (3-1/2, 4-1/2 and 8-1/2 Inch Dial Sizes).
MIL-R-19523 - Relays, Control, Naval Shipboard.
MIL-T-19646 - Thermometers, Remote Reading, Self-Indicating Dial, Gas Actuated.
MIL-T-24107 - Tube, Copper (Seamless) (Copper Numbers 102, 103, 108, 120, 122 and 142).
MIL-I-24144 - Indicators, Frost-Point/Dew-Point, Self-Cooling, Semi-Portable.
MIL-V-24578 - Valves, Globe, Pressure Instrument, Stem Test Connection, Union End.
MIL-C-52950 - Crates, Wood, Open and Covered.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-471 - Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-769 - Thermal Insulation Requirements for Machinery and Piping.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1367 - Packaging, Handling, Storage, and Transportability Program Requirements (For Systems and Equipments).
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems Electric Power, Alternating Current. (Metric)
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1629 - Procedures for Performing a Failure Mode, Effects and Criticality Analysis.
MIL-STD-1635 - Reliability Growth Testing.

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

MILITARY

803-1385850 - Piping, Instruments, Pressure for All Service.
810-2145528 - Fittings and Adaptors, Instrument, Pressure.
9000-S6202-F-73907 - Light, Indicator, Switchboard, 2 Lamp, SPF, Types B-27A Through B-27G.

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PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0967-LP-597-1011 - Electronic Equipment, Parts Application and Reliability Information Manual for Navy.
0900-LP-001-7000 - Piping System Brazed, Fabrication and Inspection.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.2 Reliability. The mean time between failures (MTBF) of the dehydrator assembly shall be not less than 3,000 hours (see 4.7.3). A failure shall be any condition or malfunction that requires the dehydrator to be taken off the line for corrective maintenance, or air output that does not meet the conditions as specified in 3.7.1.

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3.3 Maintainability. The dehydrator shall be such that all maintenance, both corrective and preventive, can be accomplished at the organizational level with no outside assistance. Preventive and corrective maintenance shall be accomplishable by a machinist mate third class with no formal equipment training as demonstrated by successful completion of the maintainability demonstration (see 4.7.4).

3.3.1 Preventive maintenance. The mean preventive maintenance time shall be not greater than 10 man-hours per month. The time required to perform any preventive maintenance action shall be not greater than 3 man-hours.

3.3.2 Corrective maintenance. The unit shall have a mean time to repair (MTTR) of not more than 3 man-hours. The maximum time to perform any corrective action shall be not greater than 5 man-hours.

3.4 General description. The dehydrator shall be a condenser-filter type and shall consist of an air to chilled water heat exchanger, air filters, condensate discharge devices with manual override, instrumentation, and controls and accessories as specified herein. The dehydrator shall be capable of effluent air flows from zero to rated flow without failure. Unless otherwise specified (see 6.2.1), the dehydrator shall be shipboard deck mounted and shall meet the performance requirements specified in 3.7.

3.4.1 Arrangement and accessibility. The dehydrator shall be configured such that operation, maintenance, and repair of the unit can be accomplished from the front and top of the unit. Piping, wiring, instruments, and all other components shall be arranged such that their partial disassembly or removal is not required to provide ready access to components requiring preventive maintenance.

3.4.2 Physical parameters. The dehydrator shall not exceed the physical parameters specified in table I. Where no size and weight limitations are provided, the physical parameters shall be as specified (see 6.2.1).

TABLE I. Dehydrator overall dimensions and weights.

Class	Capacity (ft^3/min)	Height (inches)	Width (inches)	Depth (inches)	Weight (pounds)
1	100	56 plus 24 for disassembly	30	24 plus 24 in front for disassembly	450
2	200	62 plus 34 for disassembly	30	24 plus 24 in front for disassembly	550

3.5 Materials. Materials used in the dehydrator shall meet the requirements specified herein. Where no specific requirements are identified, materials shall be corrosion-resistant in a marine atmosphere. Critical parts shall be made of corrosion-resistant materials such as corrosion-resistant steel, corrosion resistant nickel-copper-aluminum alloy in accordance with QQ-N-286, nickel-copper alloy in accordance with QQ-N-281, bronze or equivalent alloys. Other materials used in the construction of the dehydrator shall be similarly selected on the basis of corrosion-resistance and anti-galling properties. The use of cast iron shall not be permitted.

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3.5.1 Asbestos. Material used in the construction of the dehydrator package (components and supply support items) shall be asbestos-free. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

3.5.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.6 Construction. Construction shall be as specified in 3.6.1 through 3.6.7.5.

3.6.1 Human engineering. Dehydrator shall be constructed in accordance with MIL-STD-1472 to preclude or minimize the possibility of failure through improper operation or excessive and complicated maintenance procedures.

3.6.2 Inlet pressure. Construction shall be based on an inlet pressure of 150 pounds per square inch (lb/in^2) for the air and water flow paths. To avoid the need for any over-pressure protection devices, tubing and equipment furnished as part of the dehydrator shall be constructed for this pressure.

3.6.2.1 Proof pressure. The dehydrator shall withstand a pressure of 225 lb/in^2 (see 4.7.1).

3.6.3 Piping, valves, and fittings. Piping, valves and fittings shall be in accordance with MIL-STD-777, except that welded connections shall not be required.

3.6.3.1 Fittings. Pipe thread connections and bite-type fittings shall not be used. Air connections shall be O-ring gasket and retainer ring union connections. The complete union assembly shall be provided with the dehydrator for ease of installation. Piping connections shall be arranged in such a manner as to not impair the vibration-isolation properties of absorption-type mounts. Piping shall be securely supported to minimize strain and vibration.

3.6.3.2 Gauge piping. Gauge piping shall be in accordance with Drawing 803-1385850. Gauge piping shall be constructed of corrosion-resistant materials.

3.6.3.3 Valves. Valves shall incorporate a soft seating feature to ensure repeated tight shutoff. Construction of valves shall emphasize ruggedness, reliability, simplicity, and ease of maintenance. Air check valves shall not be an insert or a component of another fitting or manifold.

3.6.3.4 Threads. Threads shall conform to FED-STD-H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Components with pipe threads shall not be used. Welding, brazing, or otherwise altering a fitting which has pipe threads to render it impossible to

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take down shall also be prohibited. Bolts, nuts, studs, pins, screws, and such other fastenings or fittings shall be of a corrosion-resisting material. Screws exposed in final assembly shall be corrosion-resisting material.

3.6.4 Filters. Coalescing filters shall be installed in the air stream just downstream of the air-water heat exchanger. Unless otherwise specified (see 6.2.1), the filters shall perform with 100 percent efficiency the removal of all liquid and solid particles 3 micrometers and larger in size. Flow shall be from inside the filter element to outside. The filters shall be made of corrosion-resistant materials and shall be of the replaceable cartridge type. The filters shall drain to an associated automatic condensate discharge valve, manual drain valve and drain line (see 3.6.13). The filter shall have a dial-type differential pressure gauge (see 3.6.7.2). Maximum filter pressure drop (from clean to changeout) shall not cause the dehydrator to exceed the pressure drop requirement (see 3.7.3). Filter elements and differential pressure gauge shall withstand a pressure differential of 75 lb/in^2 without deformation or damage. The unit shall be such that a machinist mate third class can change the cartridge in one housing unassisted in less than 30 minutes using only common hand tools.

3.6.5 Interchangeability. Parts that are physically interchangeable or reversible shall be fully interchangeable or reversible with regard to performance, reliability, maintainability, function, and strength.

3.6.6 Electrical design. The dehydrator shall conform to MIL-E-917. The dehydrator shall be constructed for 115 volts alternating current (Vac), single-phase, 60 hertz (Hz), type I power, in accordance with DOD-STD-1399, section 300. Electrical and electronic components shall be derated in accordance with NAVSEA 0967-LP-597-1011.

3.6.6.1 Electrical connectors. Operational electrical connections shall be in accordance with requirement 10 of MIL-STD-454. Connectors shall be provided with mating parts. Alarm circuit connections shall be accessible in order to permit terminal connections for remote alarms and independent circuit closure contacts. The power feed line shall be designed for "hard wire" shipboard installation.

3.6.6.2 Cabling and components. Cabling and electrical components in the dehydrator shall be protected against transients from all sources. The profile of the transients to be protected against shall include a maximum voltage peak seven times normal line voltage having a duration of 5 microseconds. Transients shall not cause damage or inaccurate readings of electrical instruments exceeding the allowable tolerance for that instrument.

3.6.6.3 Wiring. Electrical cable shall be in accordance with MIL-C-915.

3.6.6.4 Terminal boards. Terminals and terminal boards shall be in accordance with MIL-C-2212.

3.6.6.5 Relays. Relays shall be in accordance with MIL-R-19523.

3.6.7 Instrumentation. Instrumentation shall be as specified in 3.6.7.1 through 3.6.7.5.

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3.6.7.1 Pressure indicators. Pressure indicators shall be in accordance with MIL-I-18997 and MIL-I-18997/1, and shall have the following characteristics:

(a) Dial size: 3-1/2 inches.
(b) Range: 0 to 200 lb/in^2.
(c) Minor graduations: 5 lb/in^2.
(d) Design: Simplex.
(e) Operating pressure pointers shall be in accordance with MIL-I-18997. One shall be set at the minimum inlet air pressure and one at the maximum (80 and 150 lb/in^2 respectively). Permanent markings on the indicator face may be substituted. The markings shall clearly designate the rated operating pressure range and the remainder of the face shall be marked as a danger zone.
(f) Indicators shall be installed in accordance with Drawings 803-1385850 and 810-2145528. Pressure gauges shall be such that they may be calibrated without removal from the instrument panel. Gauge calibration valves shall be provided in accordance with MIL-V-24578.

3.6.7.2 Differential pressure indicators. Differential pressure indicators shall be provided for all filters. The differential pressure at which filter changeout is required shall be marked on the indicating face of the differential pressure gauge or on the identification plate of the gauge. The filter gauge shall read zero under conditions of zero air flow and proportionately higher positive pressure representative of the filter media condition and air flow. Both the gauge and its identification plate shall be clearly visible to the operator. Materials and testing shall be in accordance with MIL-I-18997.

3.6.7.3 Lamps and lamp test switches. Lamp test switches shall be in accordance with MIL-S-8805 and shall be the push (snap action) type. Lamps shall be in accordance with type B27A of Drawing 9000-S6202-F-73907. Alarm lamps shall flash continuously when alarming.

3.6.7.4 Temperature indicators. Temperature indicators shall have an accuracy of plus or minus 3 percent of the full scale reading, shall be non-mercuric, and shall be in accordance with MIL-T-19646, class C. The temperature sensing bulb shall be readily accessible for gauge calibration without disassembly of the pressure vessel.

3.6.7.5 Dew point sample connection. Each dehydrator shall have a dew point sample connection installed in the dehydrator effluent air line. This sample connection shall be located in the front of the dehydrator and shall be self-sealing, quick disconnecting and fitted with a protective cap. The cap shall be attached to the instrument with a stay chain to insure against loss. This connection shall permit sampling with a dew point measuring instrument in accordance with MIL-I-24144.

3.6.8 Instrument panel. An instrument panel shall be mounted on the front of each dehydrator for centrally locating all monitoring devices. Monitoring devices shall be clearly visible to the operator while standing in front of the dehydrator. The panel shall consist of the following:

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(a) Water inlet pressure indicator.
(b) Water outlet pressure indicator.
(c) Lamp test switch.
(d) The air filter differential pressure gauges.
(e) Air outlet temperature indicator.
(f) Air outlet pressure indicator.
(g) Water inlet temperature indicator.
(h) Water outlet temperature indicator.

3.6.9 Controls. Controls shall be installed to ensure proper operation of the dehydrator. Air outlet temperatures equal to or greater than 55 degrees Fahrenheit (°F) shall actuate audible and visual alarms. An interlock shall also be installed so that the dehydrator air outlet temperature alarm shall be energized only when the associated compressor is operating. Switches used to actuate the high temperature alarms shall be in accordance with MIL-S-16032 and shall sense the air temperature within the dehydrator immediately downstream of the condenser assembly. A time delay of 10 minutes maximum shall temporarily prevent actuation of the high temperature alarms during dehydrator startup. Manual overrides shall be provided for the interlock for emergency operation or dehydrator operation with an external (shore) air source. Relays shall be in accordance with MIL-C-2212.

3.6.9.1 Audible alarms. The audible alarm shall be a buzzer in accordance with MIL-A-15303. A manual override or cutout switch shall silence the audible alarm while the visual alarm remains actuated.

3.6.10 Control panel. A control panel shall be mounted on the front of the instrument panel. The control panel shall be a watertight electrical enclosure in accordance with MIL-E-2036, and shall contain electrical terminal strips, switches, and relays. The following indicating lights and switches shall be located on the front cover of the control panel:

(a) Power "ON" lamp: A white power "ON" lamp shall light when power is supplied to the panel.
(b) High temperature alarm lamp: A red high temperature alarm lamp shall flash when the condenser effluent air exceeds 55°F.
(c) High temperature alarm lamp test switch.
(d) Alarm interlock disconnect switch: The alarm interlock disconnect switch shall function to override the compressor dehydrator alarm interlock when in the "OFF" position.
(e) Buzzer disconnect switch: The buzzer disconnect switch shall silence the audible high discharge air temperature alarm.
(f) Time delay override switch: The time delay override switch shall cancel the time delay that prevents actuation of the high temperature and compressor interlock alarm upon dehydrator and associated air compressor startup. The time delay override switch, when in the "ON" position, shall cancel the time delay circuit.

3.6.11 Air-water condenser. The air-water condenser assembly shall be chilled water cooled and shall deliver air with a dew point within 4°F of the chilled water inlet temperature under the conditions specified in 3.7. The condenser shall use a minimum fouling factor of 0.0005 hr-ft^2-°F/Btu. The condenser shall be removable from the dehydrator to allow for cleaning or replacement. Condenser tubes shall conform to MIL-T-24107 for seamless copper

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tubing or MIL-T-15005 for copper-nickel. Chilled water velocities through the tubes shall not exceed 9 feet per second. The condenser shall purge and drain both the air and the water circuits. Carbon steel shall not be used in any part of the air-water condenser.

3.6.12 Insulation. Dehydrator parts, in both the air and cooling water circuits and structural components on which condensation may form, shall be insulated in accordance with MIL-STD-769. The insulation shall be attached with removable fasteners to facilitate insulation removal during routine maintenance.

3.6.13 Condensate discharge devices. Condensate discharge devices shall be provided for the condenser, air-water condenser sump and filters. The condensate discharge device shall be constructed of corrosion-resistant materials, automatic in operation, float-actuated, pneumatic power-operated (as with a pilot-actuated valve), positive acting (as utilizing a magnetic latch or over-center levers), and shall be fitted with a manual drain valve.

3.6.14 Drains. Each drain line shall terminate at a common open funnel drain. Funnel drain shall be visible from the front of the dehydrator for verification of individual drain operation. The common drain shall terminate at the perimeter of the dehydrator with a union, ready for hard piping to the ship's drainage system. The common drain shall have a nominal pipe size of 3/8 inch.

3.6.15 Painting. Surfaces to be painted shall be thoroughly cleaned and free from oil, grease, mill scale, rust and loose weld beads. External nonworking ferrous and aluminum surfaces shall be painted with one coat of zinc chromate primer in accordance with TT-P-645 and two coats of haze-gray paint in accordance with TT-E-490.

3.6.16 Welding. Welding and allied processes shall be in accordance with MIL-STD-278. Brazing shall be in accordance with NAVSEA 0900-LP-001-7000.

3.6.17 Safety. Placement of devices requiring attention during maintenance or repair shall be such that potential danger to the operator due to high voltage, pressure, temperature, or moving parts shall be precluded.

3.7 Performance requirements. Performance requirements shall be as specified in 3.7.1 through 3.7.7.

3.7.1 Performance design requirements. Unless otherwise specified (see 6.2.1), performance design parameters for classes 1 through 3 dehydrators shall be as specified in table II.

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TABLE II. Design parameter.

Inlet air flow	
- Class 1	100 ft^3/min
- Class 2	200 ft^3/min
- Class 3	As specified (see 6.2.1)
Inlet air pressure (all classes)	80-150 lb/in^2
Inlet air temperature (all classes)	125°F (maximum)
Inlet moisture content (all classes)	0.015 lb H_2O/lb dry air (aximum)
Chilled water flow	
- Class 1	3 gal/min (maximum)
- Class 2	4 gal/min (maximum)
- Class 3	As specified (see 6.2.1)
Inlet chilled water temperature (all classes)	47°F (nominal)
Outlet air pressure (all classes)	Within 5 lb/in^2 of inlet air pressure
Outlet air dew point (all classes)	Within 4°F of inlet chilled water temperature

3.7.2 Ambient temperature. The dehydrator shall operate and conform to all performance requirements specified herein at any ambient temperature from 50 to 140°F.

3.7.3 Air pressure drop. The air pressure drop across the complete dehydrator shall not exceed 5 lb/in^2 regardless of filter condition.

3.7.4 Chilled water pressure drop. The chilled water pressure drop across the complete dehydrator shall not exceed 25 lb/in^2.

3.7.5 Shock and vibration. The dehydrator shall be resistant to mechanical shock and vibration as encountered on board ship. The dehydrator shall pass the shock and vibration tests specified in 4.7.7.

3.7.6 Inclined operation. The dehydrator shall meet all the performance requirements specified herein when permanently inclined to 15 degrees (see 4.7.6).

3.7.7 Duty cycle. The dehydrator shall operate continuously as well as intermittently from zero to rated flow.

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3.8 Identification and information plates. Identification and information plates shall be as specified in 3.8.1 through 3.8.6.

3.8.1 Location. Identification plates and information plates, where required, shall be in accordance with MIL-P-15024/5 and MIL-P-15024 respectively and shall be mechanically attached to the part of machinery or equipment that will not ordinarily be renewed during its normal service life. Plates shall be located where they will be readily visible, and secured to equipment with corrosion-resistant fasteners. Fasteners used to secure the plate shall not penetrate the frame or the end brackets.

3.8.2 Material. Plates shall be made of brass, nickel-copper alloy, corrosion-resistant steel, or anodized aluminum. Materials for label plates shall be for severe service in accordance with the requirements of MIL-P-15024/5.

3.8.3 Markings. Markings shall be either etched, stamped, engraved, metal photographed on anodized aluminum or cast in such a manner as to produce permanent markings. Etchings, engravings, or stampings shall be not greater than 0.003 inch deep. Characters on plates shall be raised to not less than 0.03 inch. Engraved, stamped or direct etched markings shall be filled with black enamel or lacquer. The identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Classification.
(d) Manufacturer's serial number.
(e) Date of manufacture.
(f) Characteristics.
    (1) Capacity.
    (2) Inlet pressure (lb/in^2).
    (3) Electrical data (for example, voltage).
    (4) Dew point at operating pressure, and temperature.
    (5) Moisture content of influent and effluent air.
    (6) Any other pertinent characteristics.
(g) Space for National stock number even though a number may not be assigned.

3.8.4 Piping connections. Inlet and outlet union ended air and cooling water connection piping and sizes shall be as specified (see 6.2.1).

3.8.5 Electrical and mechanical controls. Electrical and mechanical controls shall be marked as specified in 3.8.3.

3.8.6 Operation and safety instruction plate. An operating and safety instruction plate shall give the operator a clear, concise, step-by-step procedure for startup and operation of the dehydrator. The instruction plate shall include:

(a) Schematic flow diagram with all the dehydrator components identified therein.
(b) Safety precautions required for safe operation and maintenance.
(c) Step-by-step instructions for start up, operation, and shutdown of the dehydrator.

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(d) Applicable NAVSEA technical manual number and number of this specification.

3.9 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2).

3.10 Imaged aperture/tabulating cards. When specified in the contract or order, imaged aperture/tabulating cards shall be prepared (see 6.2.2).

3.11 Failure mode, effects, and criticality analysis (FMECA). When specified in the contract or order, a FMECA shall be prepared in accordance with MIL-STD-1629 for use during first article testing (see 6.2.2).

3.12 Failure/malfunction report, contractor engineering and technical services (CETS) report, and maintainability demonstration report. When specified in the contract or order, a failure/malfunction report, CETS report, and maintainability demonstration report shall be prepared (see 6.2.2).

3.13 Workmanship. Dehydrator units shall be constructed, assembled and finished to assure quality equipment and neat appearance that is free from imperfections that will affect durability, operability and serviceability, and safety. The metal portion of the units shall be free from fins, burrs, and sharp and rough edges.

3.14 Technical manuals. Technical manuals (including technical repair standards) shall be prepared and delivered as specified in the Technical Manual Contract Requirments (TMCR) and contract or order (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order an inspection system program plan shall be prepared (see 6.2.2).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified in the reference documents.

4.4 First article inspection and testing. The first dehydrator of each class representing production equipment in all respects shall be examined and tested by the contractor to determine conformance with the requirements of this specification. The first article inspection shall consist of the examination and tests specified herein and shall be conducted in the following order:

(a) Visual and dimensional examination (see 4.6).
(b) Hydrostatic test (see 4.7.1).
(c) Pressure decay test (see 4.7.2).
(d) Reliability test (see 4.7.3).
(e) Maintainability demonstration (see 4.7.4).
(f) Solid contaminant determination (see 4.7.5).
(g) Inclined operation (see 4.7.6).
(h) Shock and vibration (see 4.7.7).
(i) Operating test (see 4.7.8).

The contractor shall complete examination and tests (a) through (g) before completing the shock and vibration test. Upon successful completion of the shock and vibration test the dehydrator shall be subjected to the operating test (see 4.7.8). The contractor shall correct all failures, malfunctions or problems resulting from all first article test and examinations, including those performed by the Government. Failures during tests (a) through (g) will require, at the Government option, a retest by the Government. Failures during tests (h) or (i) shall require a repetition of the failed test by the contractor.

4.4.1 First article reports. When specified in the contract or order, a first article test report, notification of tests, and test procedures shall be prepared (see 6.2.2).

4.5 Quality conformance inspection. The quality conformance inspection shall be conducted as follows and shall be performed by the contractor on each dehydrator. Shutdown, alteration to equipment or adjustment of any kind prior to completion of each test shall void the inspection unless specifically required by the inspection procedure. Order of tests may be designated by the contractor:

(a) Visual and dimensional examination (see 4.6).
(b) Hydrostatic test (see 4.7.1).
(c) Pressure decay (see 4.7.2).
(d) Operating test (see 4.7.8).

4.5.1 Test procedures. When specified in the contract or order, quality conformance test procedures shall be prepared (see 6.2.2).

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4.6 Visual and dimensional examination. Each dehydrator shall be examined to determine conformance to this specification. Equipment shall be surface-examined including visual examination for defects, workmanship, dimensions, and any other requirements not involving tests.

4.7 Tests and demonstrations. Tests and demonstrations shall be as specified in 4.7.1 through 4.7.7.

4.7.1 Hydrostatic test. The air and water flow paths of the dehydrator shall be tested hydrostatically with clean fresh water at the applicable pressure (see 3.6.2.1) for 1/2 hour. The chilled water cooling circuit shall be tested without air pressure. The air circuit shall be tested without chilled water pressure. Any leakage, porosity, or permanent deformation shall be cause for rejection.

4.7.2 Pressure decay test. The air side of each dehydrator shall be tested pneumatically for pressure drop at nominal operating pressure (no flow through dehydrator) for 1 hour. Assuming temperature corrections measured by a surface measuring instrument, the allowable pressure drop shall not exceed 1 lb/in^2 during the 1-hour test period.

4.7.3 Reliability test. A reliability development growth test (RDGT) shall be conducted for 3000 hours in accordance with MIL-STD-1635 to ensure that the dehydrator will meet the requirements of 3.2. The dehydrator shall be operated at maximum rated flow with saturated influent air at the maximum operating temperature and minimum operating pressure under maximum ambient conditions (see 3.7.2). The test shall be conducted by the contractor and the contractor shall determine any immediate and permanent corrective actions for each problem, malfunction and failure. The effectiveness of each corrective action shall then be demonstrated under the same conditions that existed when the original failure occurred. Fresh chilled water shall be used when testing the condenser-filters. Calibrated flow meters, with an error not exceeding 5 percent, shall be used. The calibration data, name of manufacturer, and model number for the flow meters shall be recorded. Dew point measurements of the effluent air shall be recorded hourly to establish dehydrator performance. Dew point error shall not exceed 2°F. Specified flow rate shall be maintained within minus 3 percent and plus 5 percent throughout the test. Performance testing shall be conducted with ambient temperature, flow rate, inlet air temperature, and inlet air pressure as specified (see 3.7.1) by the individual test.

4.7.4 Maintainability demonstration. A maintainability demonstration shall be conducted in accordance with MIL-STD-471, phase II. The demonstration shall be performed to ensure conformance to the maintainability requirements (see 3.3).

4.7.5 Solid contaminant determination. Independent tests performed on the filter cartridge may be substituted for this test procedure. Should a filter be used which has been successfully tested for this specification in the past, a request for waiver of testing based upon prior acceptance may be submitted. Dehydrators shall meet the requirements specified in 3.6.4.

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4.7.6 Inclined operation. The dehydrator shall be operated at rated load for a minimum of 1 hour. During this period the base shall be tilted 15 degrees for 15 minutes in each of the four positions; right, left, front and rear. During the test, there shall be no degradation of effluent air quality and all drains shall operate correctly.

4.7.7 Shock and vibration. Upon successful completion of the tests specified in 4.7.1 through 4.7.6, the dehydrator shall be subjected to the type A, high-impact shock tests specified for grade A, class I equipment in accordance with MIL-S-901, and for resistance to vibration in accordance with type I of MIL-STD-167-1. During these tests, the unit shall be energized and the air and water sides pressurized to the nominal operating pressure. When specified in the contract or order, high-impact shock test procedures and a vibration testing report shall be prepared (see 6.2.2).

4.7.8 Operating test. The first dehydrator produced shall be operated at maximum rated flow for 8 hours, with inlet air at the maximum operating temperature and minimum operating pressure (see 3.7.1) and under maximum ambient conditions (see 3.7.2). All follow units shall be tested under the same conditions for a period of 1 hour. Any operational malfunction of the dehydrator components, including any malfunction of the automatic condensate discharge device, shall be cause for rejection.

4.8 Technical manuals. Technical manual quality assurance management data shall be prepared and delivered as specified in the TMCR and contract or order (see 6.2.2).

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2.1).

5.1.1 General.

5.1.1.1 Preservation and packing data. When specified in the contract or order, preservation and packing data shall be prepared (see 6.2.2).

5.1.1.1.1 Packing, handling, storage and transportability (PHST). When specified (see 6.2.1), the system or equipment program or acquisitioning manager shall establish the applicable PHST requirements. The plan shall be tailored in accordance with MIL-STD-1367.

5.1.1.2 Lubrication. Rotating joints, bearings, and similar moving parts shall be lubricated thoroughly with the lubricant required for service use. Excess lubricants shall be removed prior to packaging.

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5.1.1.3 Painting. Painted surfaces on which the paint is damaged or defective shall be cleaned and repainted with the original specified paint of the same quality and color (see 3.6.15).

5.1.1.4 Cushioning and wrapping materials. The use of excelsior, newspaper, shredded paper (all types) and similar hydroscopic or nonneutral materials, and all types of loose fill materials for applications such as cushioning, fill, stuffing, and dunnage shall be prohibited. Materials selected for cushioning and wrapping shall be fire-resistant.

5.1.1.5 Asbestos. Asbestos or material and items containing asbestos shall not be used in the packaging of material covered by this specification. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

5.1.2 Level A. Each dehydrator unit shall be cleaned, dried, and unit protected in accordance with method I of MIL-P-116 and as follows:

(a) Exposed metal surfaces subject to corrosion shall be preserved with type P-19 preservative, specified in MIL-P-116.
(b) Internal surfaces or parts normally in contact with water shall be preserved with type P-21 preservative, specified in MIL-P-116.
(c) Openings shall be sealed with pipe plugs, plastic caps or plugs, or pressure sensitive, cloth-backed, waterproof tape.
(d) Throwaway or replaceable filter elements shall be replaced with new filters meeting the requirements of this specification.

5.1.3 Level C. Each dehydrator unit shall be cleaned, dried, and preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2.1).

5.2.1 Level A. Each unit packaged as specified (see 5.1), shall be packed in a container conforming to any one of the following specifications at the option of the contractor:

Specification	Container	Classification
PPP-B-601	Box, wood, cleated plywood	Overseas
PPP-B-621	Box, wood, nailed and lock-corner	Overseas
MIL-C-52950	Crate, wood, open and covered	Covered, style A or B, type V, demountable
MIL-C-104	Crate, wood, lumber and plywood sheathed, nailed and bolted	Type II, class 2

Container closure and reinforcement shall be in accordance with the applicable container specification or appendix thereto. Anchoring, blocking, bracing, and cushioning (see 5.1.1.4) shall be in accordance with MIL-STD-1186. Wood boxes

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shall not exceed the weight limitations of the applicable box specification. Boxes exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-C-104.

5.2.2 Level B. Packing shall be as specified for level A (see 5.2.1), except wood and plywood boxes shall be of the domestic type and waterproofing shall not be required.

5.2.3 Level C. Each dehydrator shall be packed for shipment in accordance with ASTM D 3951.

5.2.3.1 Container modification. Containers exceeding 200 pounds gross weight shall have a minimum of two, 3- by 4-inch nominal wood skids, laid flat, which shall support the item and facilitate handling by mechanical handling equipment during transportation, storage and stowage. Containers exceeding 200 pounds gross weight shall be modified in accordance with MIL-C-104.

5.3 Technical manuals. Two copies of each of the current operation/ maintenance technical manuals shall be packaged with each sytem or equipment. Technical manual packaging shall be as specified (see 6.2.1).

5.4 Marking. In addition to any special marking required (see 6.2.1 or herein), level A, B and C shipment markings, including bar coding, shall be in accordance with MIL-STD-129.

5.4.1 Special markings. When specified (see 6.2.1), the following markings in accordance with MIL-STD-129 and the applicable container specification or appendix thereto shall be made: structural markings, handling markings such as center of balance, sling and lift points, load bearing and fork lift areas and arrows.

5.5 Unpacking instructions. Unpacking instructions in accordance with MIL-STD-129 shall be placed in a plastic, waterproof bag and sealed. The bag shall be prominently marked "UNPACKING INSTRUCTIONS" and firmly affixed to the outside of the shipping container in a protected location (preferably between the cleats on the container end adjacent to the identification marking).

6. NOTES

6.1 Intended use. Dehydrators covered by this specification are intended to remove particulate matter, aerosols and water vapor from influent low pressure air and to furnish clean, dry air for use in the low pressure air system.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2.1).
(c) First article testing, when required (see 3.1).
(d) Mounting, if other than specified (see 3.4).

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(e) Physical parameters, if other than specified (see 3.4.2).
(f) Air and chilled water inlet and outlet design conditions for class 3 dehydrators other than those specified (see 3.7.1).
(1) Outlet particle size (micrometer rating (see 3.6.4)).
(2) Outlet air flow rate.
(g) Air and water inlet and outlet piping: physical location on dehydrator exterior surface (see 3.8.4).
(h) Whether technical manuals are required (see 5.3).
(i) Inspection conditions, if other than as specified (see 4.3).
(j) Level of preservation and packing required (see 5.1 and 5.2).
(k) PHST plan, if applicable (see 5.1.1.1.1).
(l) Special markings, if required (see 5.4.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Technical manuals (including Technical Repair Standards (TRS) shall be acquired under separate CLINs or subCLINs. Technical manual products (manuscript copy, preliminary copy, reproducible copy, and so forth) shall be developed and delivered as specified in the Technical Manual Contract Requirements (TMCR) exhibits to the contract. CDRLs and DIDs shall not be used for technical manual product acquisitions. Technical manual management data (plans, reports, and so forth) shall be acquired under separate data CLIN, but delivery shall continue to be specified on CDRLs and DIDs. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.5.1 and 5.1.1.5	Certificate of compliance	DI-E-2121	----
3.9	Drawings, engineering and associated lists	DI-E-7031	Level 3
3.10	Imaged aperture/tabulating cards	DI-E-20477	----
3.11	Failure mode, effects, and criticality analysis report	DI-R-7085	----
3.12	Report, failure/malfunction	UDI-T-23724	----
3.12	Report, contractor engineering and technical services (CETS)	UDI-E-23127	----
3.12	Report, maintainability demonstration	DI-R-7113	----
4.1.2	Inspection system program plan	DI-R-4803	----
4.4.1	Notification of tests	DI-T-23731	----
4.4.1	Report, first article test	UDI-T-23790	----

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Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4.1 and 4.5.1	Procedures, test	UDI-T-23732	----
4.7.7	Procedures, high-impact shock test	DI-T-23757	----
4.7.7	Report, vibration testing	UDI-T-23762	----
5.1.1.1	Preservation and packing data	DI-PACK-80120	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any taks in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Definition. Technical Manuals Contract Requirements (TMCR) is an exhibit or attachment to a contract specifying technical manual product development and delivery requirements or management data development requirements.

6.4 First article. When a first article inspection is required, the item should be a first article item. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions. The following definitions are included to provide a common basis for dehydrator design and to ensure that the requirements specified herein are interpreted properly:

(a) Dew point or frost point. Dew point or frost point is the temperature at which a vapor begins to deposit as a liquid or frost.
(b) Saturated air. Saturated air is air which, at a given pressure and temperature, contains as much moisture as it can retain in the presence of an excess of moisture.

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(c) Aerosol. Aerosol is a disperson of solid or liquid particles of microscopic size in air.

(d) Maximum load. Maximum load is the set of operational conditions including maximum ambient temperature (120°F), maximum cooling water temperature (47°F), minimum inlet pressure (80 lb/in^2), nominal air flow rate (see 1.2.1), and maximum inlet air moisture content (saturated at 125°F, that is 0.015 lb H_2O/lb dry air for dehydrators).

(e) Minimum nominal load. Minimum nominal load is the set of operational conditions including ambient temperature (50°F), minimum cooling water temperature (47°F), maximum air inlet pressure (150 lb/in^2), zero air outlet flow, and nominal inlet air moisture content of 0.003 lb H_2O/lb air (saturated at 85°F). For production test purposes, ambient environmental temperature may be used.

(f) Automatic. Automatic implies that no equipment adjustment or manual operations are required for dehydrator startup, operation, or shutdown except the electrical connection and disconnection (by means of an appropriate switch) at startup and shutdown, respectively. Necessary air system piping alignment is not considered as part of the dehydrator operation for the above definition of automatic.

(g) Mean-time-to-repair. Mean-time-to-repair is the sum of corrective maintenance times at any specific level of repair, divided by the total number of failures within an item repaired at each level, during a particular interval under stated conditions.

(h) Preventive maintenance. Preventive maintenance is all actions performed in an attempt to retain an item in specified condition by providing systematic inspection, detection, and prevention of incipient failures.

(i) Corrective maintenance. Corrective maintenance is all actions performed as a result of failure, to restore an item to a specified condition. Corrective maintenance can include any or all of the following steps: localization, isolation, disassembly, interchange, reassembly, alignment and checkout.

(j) Standard cubic feet per minute (ft^3/min). Standard conditions for specified airflows are as follows: 68°F, 36 percent relative humidity (RH), and 14.7 lb/in^2 absolute.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.8 Subject term (key word) listing.

Air-water condenser
Coalescing filter
Effluent
Influent

Preparing activity:
Navy - SH
(Project 4440-N049)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-24693(SH)	DEHYDRATOR, CONDENSER-FILTER LOW PRESSURE AIR, FOR SPECIAL APPLICATIONS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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